Scientific Skills Exercise

Describing and Interpreting Quantitative Data

How Do Desert Mice Maintain Osmotic Homeostasis? The sandy inland mouse (now known scientifically as *Pseudomys hermannsburgensis*) is an Australian desert mammal that can survive indefinitely on a diet of dried seeds without drinking water. To study this species' adaptations to its arid environment, researchers conducted a laboratory experiment in which they controlled access to water. In this exercise, you will analyze some of the data from the experiment.

How the Experiment Was Done Nine captured mice were kept in an environmentally controlled room and given birdseed (10% water by weight) to eat. In part A of the study, the mice had unlimited access to tap water for drinking; in part B of the study, the mice were not given any drinking water for 35 days, similar to conditions in their natural habitat. At the end of parts A and B, the researchers measured the osmolarity and urea concentration of the urine and blood of each mouse. The mice were also weighed three times a week.

Data from the Experiment

	Mean Osmolarity (mOsm/L)		Mean Urea Concentration (m*M*)	
Access to Water	Urine	Blood	Urine	Blood
Part A: Unlimited	490	350	330	7.6
Part B: None	4,700	320	2,700	11

In part A, the mice drank about 33% of their body weight each day. The change in body weight during the study was negligible for all mice.

INTERPRET THE DATA

1. In words, describe how the data differ between the unlimited water and no-water conditions for the following: **(a)** osmolarity of urine, **(b)** osmolarity of blood, **(c)** urea concentration in urine, **(d)** urea concentration in blood. **(e)** Does this data set provide evidence of homeostatic regulation? Explain.
2. **(a)** Calculate the ratio of urine osmolarity to blood osmolarity for mice with unlimited access to water. **(b)** Calculate this ratio for mice with no access to water. **(c)** What conclusion can you draw from these ratios?
3. If you learned that the amount of urine produced in part A were different from that in part B, how would that affect your calculation? Explain.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Data from R. E. MacMillen et al., Water economy and energy metabolism of the sandy inland mouse, *Leggadina hermannsburgensis, Journal of Mammalogy* 53:529–539 (1972).

Transport Epithelia in Osmoregulation

Although the ultimate function of osmoregulation is to control solute concentrations in cells, most animals do this indirectly, managing instead the solute content of the body fluid that bathes the cells. In insects and other animals with an open circulatory system, the fluid surrounding cells is hemolymph. In vertebrates and other animals with a closed circulatory system, the cells are bathed in an interstitial fluid that contains a mixture of solutes controlled indirectly by the blood. Maintaining the composition of such fluids depends on structures ranging from individual cells that regulate solute movement to complex organs such as the vertebrate kidney.

In most animals, osmoregulation and metabolic waste disposal rely on **transport epithelia**—one or more layers of epithelial cells specialized for moving particular solutes in controlled amounts in specific directions. Transport epithelia are typically arranged into tubular networks with extensive surface areas. Some transport epithelia face the outside environment directly, whereas others line channels connected to the outside by an opening on the body surface.

The transport epithelium that enables the albatross and other marine birds to survive on seawater remained undiscovered for many years. To find it, researchers gave captive marine birds only seawater to drink. Although very little salt appeared in the birds' urine, tests revealed that fluid dripping from the tip of their beaks was a concentrated solution of salt (NaCl). The source of this solution turned out to be a pair of nasal salt glands packed with transport epithelia **(Figure 44.5)**. Salt glands, which are also found in sea turtles and marine iguanas, use active transport of ions to secrete a fluid much saltier than the ocean. Even though drinking seawater brings in a lot of salt, the salt gland enables these marine vertebrates to achieve a net gain of water. By contrast, humans who drink a given volume of seawater must use a *greater* volume of water to excrete the salt load, with the result that they become dehydrated.

Transport epithelia that function in maintaining water balance also often function in disposal of metabolic wastes. We'll see examples of this coordinated function in our upcoming consideration of earthworm and insect excretory systems as well as the vertebrate kidney.